# GONZALO LOAYZA

Provo, UT • +1 (801) 735-8034 • gloayza5@byu.edu • LinkedIn • Portfolio

## SUMMARY

Senior Computer Science student at BYU with a Machine Learning emphasis and hands-on experience in data science, applied analytics, and ML workflows using real-world operational datasets. Skilled in Python, SQL, time-series analysis, feature engineering, model evaluation, and decision-support analytics. Strong ability to frame business and operational problems, identify value drivers, and apply machine learning to support practical decisions in industrial, infrastructure, and environmental contexts.

## EDUCATION

### Brigham Young University (BYU), Provo, UT

B.S. in Computer Science, Machine Learning Emphasis

Expected Graduation: December 2026

Relevant Coursework: Deep Learning, Data Science Capstone, Machine Learning, Algorithms, Data Structures, Computer Systems, Probability & Statistics

Honors Thesis: Operational Mode Discovery and Business Value Analysis in Industrial Time-Series Data

## EXPERIENCE

### Data Analytics Intern | Hatch Digital, Peru

June 2026 - August 2026

- Contributed across three applied digital workstreams: tailings-data validation and automation, drone-enabled measurement concepts, and shift-level decision-support workflows for mining and engineering applications.
- Supported problem framing, data preparation, operational KPI logic, and multidisciplinary validation so analytical outputs remained connected to real engineering decisions.
- Presented the internship work and lessons learned to the Hatch Young Professionals community through a technical retrospective focused on measurement, data reliability, and operational decision support.

### Maintenance Data Analyst Intern | Compañía de Minas Buenaventura, San Gabriel Unit, Peru

Jun 2025 - Aug 2025

- Identified reliability and maintenance-analysis gaps in crushing and grinding equipment data, then processed and cleaned 50k+ noisy sensor time-series to support predictive-maintenance exploration.
- Analyzed operating signals linked to equipment wear, ore variability, and process behavior to support condition-monitoring and reliability-focused decision-making.
- Engineered 15+ features, including temperature deltas, load ratios, and transient-spike indicators, to improve the analytical basis for equipment monitoring workflows.
- Structured 100+ assets under ISO 14224/17359 standards, improving maintenance-data traceability, consistency, and analytical usability.

### Data Analytics Intern | Hatch Ltd, Urban Solutions Sector, Vermont, USA

May 2024 - Aug 2024

- Evaluated operational delay and network-performance challenges across simulation scenarios, then analyzed 200k+ records to identify bottlenecks and support planning decisions.
- Built Python and C++ automation workflows that reduced processing time by about 70%, improving analysis speed, repeatability, and scenario-evaluation capacity.
- Integrated rider-survey information with onboard sensor data to strengthen scenario interpretation and validate service-level assumptions.
- Supported planning teams through statistical and optimization-based analysis, translating simulation outputs into practical insights for operational decision-making.

## TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript

Data Science & ML: time-series analysis, feature engineering, classification, unsupervised learning, model evaluation, exploratory data analysis, dimensionality reduction, predictive analytics

Libraries & Tools: scikit-learn, Pandas, NumPy, Jupyter, Git, Linux, SHAP, Power BI

Business & Industrial Analytics: problem framing, KPI analysis, bottleneck analysis, value-driver identification, operational mode discovery, dashboard-oriented thinking, decision-support analytics

## PROJECTS

### Honors Thesis - Operational Mode Discovery and Business Value Analysis in Industrial Time-Series Data

Python | Time-Series Analytics | PCA | DBSCAN | Business Value Analysis - BYU Honors Program, 2025 - 2026

- Framed an industrial analytics problem around operating variability, recovery losses, throughput pressure, and equipment bottlenecks in a real processing environment.
- Integrated minute-level process data with daily KPI context, standardized process features, and reduced dimensionality with PCA using a 95% explained-variance target.
- Applied DBSCAN and identified three recurrent operating regimes, then profiled them against recovery, production, variability, and equipment-configuration signatures.
- Connected data infrastructure, control-loop performance, instrumentation gaps, and ML opportunities to a staged roadmap focused on throughput, recovery stability, energy optimization, and business value.

### Wildfire Prediction System

Python | XGBoost | Time-Series | API Integration | Docker - BYU, 2026

- Built a wildfire-risk prioritization prototype using 7-day weather histories and reported Utah fire data to narrow where satellite-image follow-up should occur first.
- Trained and compared XGBoost, Random Forest, and Naive Bayes models on balanced and imbalanced wildfire-event datasets.
- Selected the unbalanced XGBoost workflow, which achieved 23% precision, 9.7% recall, and PR AUC 0.110 versus a 0.006 random baseline on the evaluated rare-event task.
- Connected model scoring to a browser-based risk map and satellite-image retrieval workflow using a Python backend and containerized services.

## CERTIFICATIONS

AWS Certification in Progress

## AWARDS

Donald Goodyear Doll Sr. Scholarship | Dr. Gerald Hatch Scholarship | BYU Honors Program

## LEADERSHIP

Emergency Response & Rescue Program, Peruvian Army
Leadership training focused on coordination, rapid decision-making, and execution in high-pressure environments.

## LANGUAGES

English (Fluent) | Spanish (Native) | French (Intermediate)